Optimization and AI for University Course Timetabling

STATE-OF-THE-ART REVIEW AND A REPRODUCIBLE COMPARISON OF CP-SAT AND SIMULATED ANNEALING

University course timetabling sits between two research traditions. Operations Research formulates the problem through integer or constraint models; Artificial Intelligence contributes heuristic search, hyper-heuristics, and, more recently, learning-assisted decisions. This project will review both traditions and then test one solver from each under identical conditions. The aim is a fair comparison, not another algorithm whose advantage depends on a different dataset or time limit.

RATIONALE AND RESEARCH QUESTIONS

The University Course Timetabling Problem (UCTP) is NP-hard. Metaheuristics remain common [1], but exact and matheuristic models are also active lines of research [2,5]. Machine Learning appears mainly in supporting roles, such as algorithm selection, parameter control, and surrogate evaluation, rather than as a complete timetable generator [2,3]. A review restricted to Genetic Algorithms and Simulated Annealing would therefore miss a substantial part of the current field.

The study asks two questions. RQ1: What evidence supports exact, metaheuristic, matheuristic, and learning-assisted approaches, and how reproducible are their comparisons? RQ2: Under a common time budget, how do CP-SAT and Simulated Annealing (SA) differ in feasibility, timetable quality, robustness, and ability to quantify solution quality? We expect CP-SAT to provide useful bounds and strong solutions on smaller instances, while SA may behave more consistently as search spaces grow.

RESEARCH DESIGN

PHASE 1	Systematic search. We will search Scopus, Web of Science, IEEE Xplore, ACM Digital Library, and ScienceDirect for peer-reviewed studies published between 2015 and 2026. Earlier foundational work will be recovered by backward citation searching. Queries will combine university course timetabling with terms for integer and constraint programming, metaheuristics, matheuristics, hyper-heuristics, Machine Learning, and benchmark evaluation. Selection and exclusions will be recorded in a PRISMA-style flow diagram.
PHASE 2	Critical synthesis. Studies will be grouped as exact, metaheuristic, matheuristic/hybrid, or learning-assisted. The extraction matrix will record the problem variant, dataset, constraints, objective, tuning, stopping rule, baselines, statistical analysis, code and data availability, and stated limitations. A quality checklist will separate reproducible comparisons from results that cannot be independently verified. The role of ML will be assessed by function, not by whether a paper labels its method as AI.
PHASE 3	Implementation. CP-SAT will encode the hard constraints and the official weighted soft objective. It can return feasible schedules, objective bounds, and optimality certificates when time permits. SA will start from a constructive schedule and use move/swap neighborhoods, geometric cooling, reheating, and repair. A greedy constructor will provide a baseline. All three methods will use the same parser, validator, objective calculation, and output format.

EXPERIMENTAL PROTOCOL

The experiment will use the 21 public curriculum-based instances from ITC-2007 Track 3, built from University of Udine data [4]. The official formulation separates hard feasibility from four soft penalties: room capacity, minimum working days, curriculum compactness, and room stability. Parameters will be tuned on a reserved subset and frozen before testing. Final runs will use two predeclared time limits; stochastic configurations will be repeated with at least ten seeds per instance.

Reported outcomes will include feasibility rate, official penalty, time to first feasible and best solution, CP-SAT bound, relative gap to the best-known score, and dispersion across runs. Paired comparisons will use confidence intervals, a Wilcoxon test, and an effect-size measure. Per-instance tables and convergence plots will show where each method succeeds or fails; infeasible and timed-out runs will remain in the analysis.

EXPECTED CONTRIBUTION, FEASIBILITY, AND LIMITS

The final paper will connect the claims found in the literature with evidence from a controlled benchmark. The practical output will be a documented implementation with fixed seeds, parameter files, and machine-readable results. ML will be discussed where it adds measurable value; training a decorative model on only 21 instances is outside the scope. ITC-2007 is public and anonymized, so no personal data are processed. The discussion will address parameter sensitivity, stochastic variation, hardware-dependent timing, and transfer from benchmark instances to live university scheduling.

REFERENCES

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